Experiment: 24-Hour Digital Clock Using FSM

Objective:
To design a 24-hour digital clock using flip-flops and synchronous counters.

Theory:
The digital clock consists of three counters: seconds, minutes and hours. Seconds and minutes count from 00 to 59, while hours count from 00 to 23. Carry signals are used to increment the next counter when the previous counter reaches its maximum value.

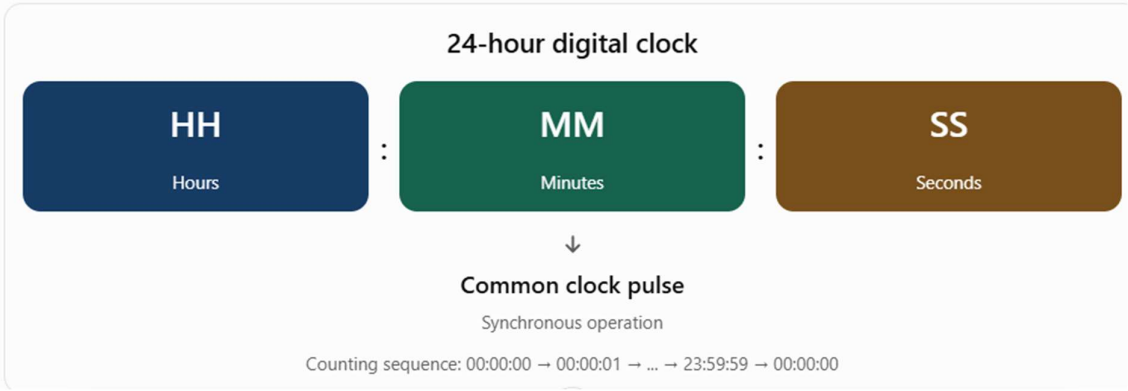
Working:
The seconds counter increments with every clock pulse. At 59, it resets to 00 and increments the minutes counter. Similarly, minutes increment hours at 59. After 23:59:59, the clock resets to 00:00:00.

C. State Description

The complete clock can be divided into three interconnected counters.

Counter	States	Next action
Seconds	00–59	Increment minutes at 59
Minutes	00–59	Increment hours at 59
Hours	00–23	Reset to 00 at 23

The complete clock has the following counting sequence:



E. State Transition Table

Here, `CLK` represents the common clock pulse.

Current time	Input condition	Next time
00:00:00	CLK = 1	00:00:01
00:00:01	CLK = 1	00:00:02
00:00:58	CLK = 1	00:00:59
00:00:59	CLK = 1	00:01:00
00:59:59	CLK = 1	01:00:00
12:59:59	CLK = 1	13:00:00
22:59:59	CLK = 1	23:00:00
23:59:58	CLK = 1	23:59:59
23:59:59	CLK = 1	00:00:00

For `CLK = 0`, the current state remains unchanged.

2. Flip-flop input equations

For a BCD digit with four D flip-flops, let its present bits be Q_3, Q_2, Q_1, Q_0.

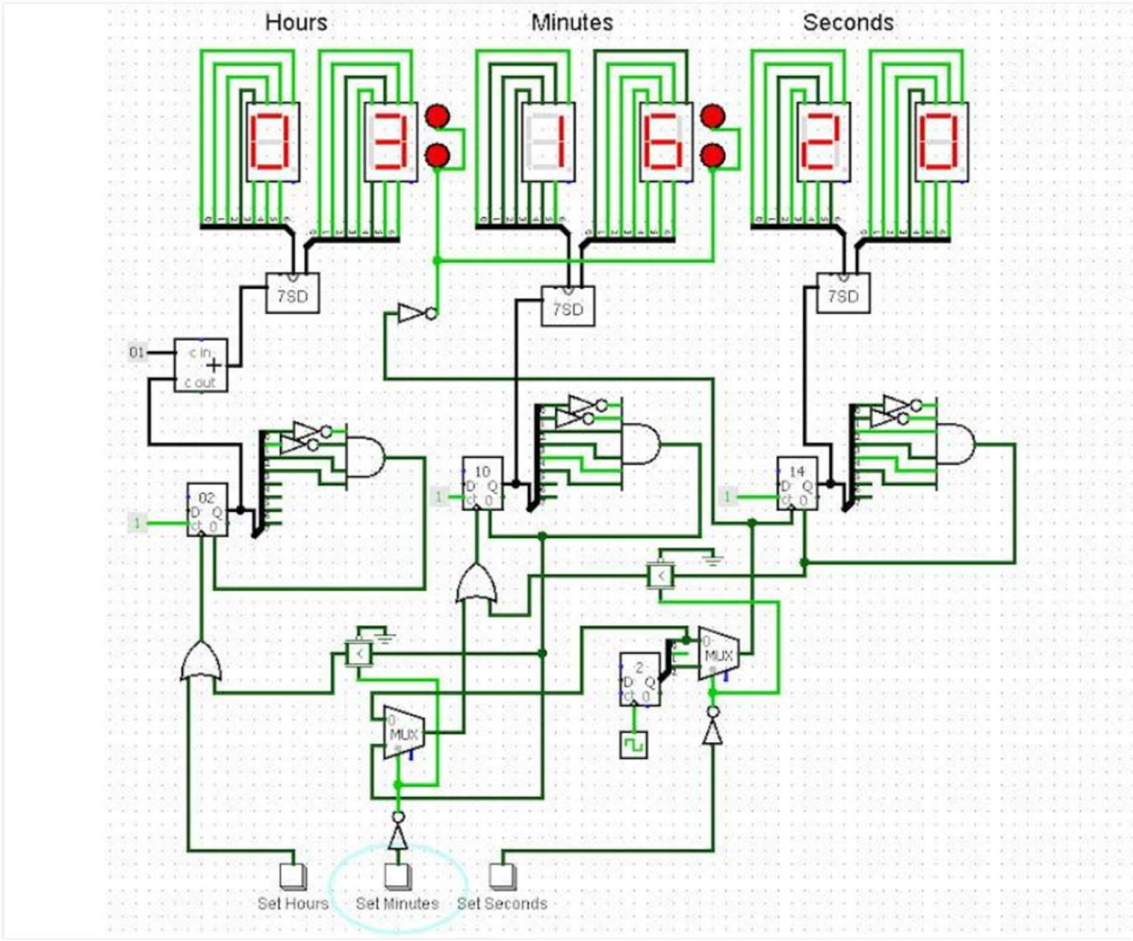
For an enabled normal increment, the binary increment equations are:

$$D_0 = \overline{Q_0}$$

$$D_1 = Q_1 \oplus Q_0$$

$$D_2 = Q_2 \oplus (Q_1 Q_0)$$

$$D_3 = Q_3 \oplus (Q_2 Q_1 Q_0)$$



Result:
A 24-hour digital clock was successfully designed using synchronous counters and flip-flops.

Conclusion:
The circuit correctly performs counting and rollover operations in HH:MM format.

https://chatgpt.com/share/6ae49ee8-6e7c-83ee-a64e-f817ad1ecd6d